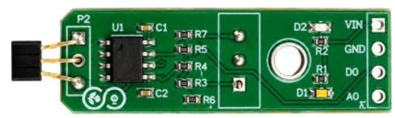
LINEAR MAGNETIC HALL SENSOR **Datasheet**

The Linear Magnetic Hall Sensor Module detects magnetic fields using the Hall effect principle, producing a digital output signal when a magnetic field is sensed. It features a built-in LED indicator to visually represent the detection status. The sensor includes three digital interfaces for easy integration and allows for sensitivity adjustments via a potentiometer. The output is a digital voltage signal proportional to the magnetic induction intensity.

Key Features

- Features wide range voltage comparator LM393
- Adjustable sensitivity
- Signal output indicator with the retaining bolt hole, convenient installation
- Output form digital switch output (0 and 1 high and low level)
- Signal output indication
- Single channel signal output
- The output effective signal is low level
- When there is sound, outputs low level and the signal light
- Can be used for Acoustic control light, give sound and light alarm working with the Photosensitive sensor, and sound control, sound detect



Hall Sensor Board

Technical Specifications

Parameter	Value
Supply Voltage (Vcc)	3.3 V to 24 V
Magnetic Range	±50 mT to ±1000 mT
Output Types	Analog: Voltage proportional to magnetic field strength Digital: Binary on/off output PWM: Duty cycle proportional to field strength
Current Consumption	Few mA to tens of mA
Sensitivity	1 to 100 mV/mT
Temperature Range	−40 °C to +150 °C
Response Time	Microseconds to milliseconds
Hysteresis	Few % of magnetic range
Linearity	Defined as % of full-scale output

Application

- **Automotive Systems —** Wheel speed sensing, gear position detection, pedal position monitoring.
- **Motor Control —** Rotor position detection in BLDC and stepper motors.
- **Consumer Electronics —** Open/close detection in phones, laptops, and smart appliances.
- **Industrial Automation —** Proximity sensing, machine safety interlocks, and current sensing.

- **Medical Devices —** Position and movement detection in diagnostic or monitoring equipment.